

Mohamed Keshk

Mechatronics Engineer

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Professional Summary

Senior Mechatronics Engineer with 9 years of hands-on experience designing, prototyping, and refining complex robotic systems and autonomous infrastructure. Proven expertise in designing mechanical assemblies, running FEA simulations, and seamless mechatronic integration of sensors and motors. Adept at leveraging lightweight, high-strength materials (carbon fiber, aluminum) and utilizing ROS for hardware-software testing. Strong track record of leading product life-cycles from rapid prototyping to series production in cross-functional, international environments.

Professional Experience

Avemoy FlexCo

Feb 2025 – Present

Mechatronics Team Lead, Austria

Lead the mechatronics and mechanical development of autonomous fleets and automated ground support equipment, focusing on precision, durability, and scalability.

- **Rapid Prototyping & Integration (The Dragonfly):** Designed mechanical assemblies and mounting structures for 7-inch testing platforms. Integrated depth cameras, LiDAR, and high-torque motors to create a seamless mechatronic testing environment.
- **Hardware-Software Testing:** Executed 300+ field tests using ROS and custom testing scripts to validate sensor integration, motion systems, and firmware features.
- **Series Production (The Firefly):** Directed the full mechanical and hardware life-cycle for a lightweight 4-inch autonomous drone. Utilized carbon fiber to achieve a lightweight yet high-strength build.
- **Mechanical Ground Systems:** Designed and engineered automated Ground Support Equipment (GSE) featuring precision localization features for autonomous landing, charging, and data offload.

Innovation Hub - Nile University

Sep 2022 – Feb 2025

Lead R&D Engineer, Egypt

- Acted as technical lead for researchers and robotics startups, providing prototyping and production consultation, design-for-manufacturing (DFM) reviews and tolerance analyses for complex mechatronic systems.
- **Robotic Actuation & Fluid Dynamics:** Designed and manufactured mechanical assemblies for a hubless rim-driven brushless DC thruster utilized in autonomous underwater vehicles (AUVs).
- **Rapid Prototyping:** Built custom brushless motor winding machines incorporating gears, linkages, and precise motion systems to accelerate the prototyping of electromagnetic actuators.
- Collaborated across software and hardware teams to ensure seamless integration of custom embedded systems and mechanical exoskeletons/frames for 25+ advanced graduation projects.

Innovation Hub - Nile University

Sep 2021 – Sep 2022

R&D Lab Engineer, Egypt

- Maintained rapid prototyping in-house tools (CNC, 3D printers, manual machining) and provided technical DFM support for engineering teams.
- Taught and facilitated ECEN labs focusing on Physical Sensors, Transducers, and Analog/Digital Electronics.

Tensor Electronics

Jul 2015 – Sep 2021

Electronic Design Engineer, Egypt

- Designed smart-home and automation system modules from schematic capture through to mass production.
- Developed hardware and firmware for RF communication modules, ensuring reliable data transfer between structural nodes and central systems.

Education

- **Robotics Project (AMR):** Developed and engineered an Autonomous Mobile Robot (AMR) with a 200 kg payload capacity. Designed structural mechanical assemblies and utilized ROS and LiDAR SLAM for navigation and hardware-software testing.
- **Robotic Actuation Research:** Designed a neural-network-controlled Stewart Platform (6-DOF) utilizing complex linkages, motion systems, and joint actuation. Ran kinematics, FEA, and tolerance analyses using MATLAB and MSC Adams co-simulation.

Technical Skills

- **Mechanical Design & CAD:** SolidWorks (Advanced - CSWA Mechanical Design & Additive Manufacturing), Fusion 360, Design for Manufacturing/Assembly (DFM/DFA).
- **Mechatronics & Robotics:** Robotic joints, exoskeletons, motion systems, gears, linkages, actuator integration, sensor integration, hardware-software testing.
- **Analysis & Prototyping:** FEA simulations, tolerance analyses, MSC Adams, Ansys, 3D printing (SLA/SLS/FDM), CNC machining, carbon fiber & aluminum fabrication.
- **Software & Control:** ROS, Python, C/C++, MATLAB, embedded microcontrollers (ESP32, STM32).
- **Electronics:** PCB design (KiCad), mixed-signal circuits, power electronics, communication protocols (CAN, I2C, SPI).